

PETER ZHANG

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Highly motivated Ph.D. candidate in Computational Chemistry with expertise in RNA structural biology and computational biophysics, including atomistic and coarse-grained molecular dynamics simulations and force-field development. Experienced in integrating experimental data with computational modeling to address complex challenges in biomedical research and drug discovery. Published in peer-reviewed journals and proficient in Python, high-performance computing, and GPU-accelerated scientific workflows. Seeking to apply these skills to oncology drug discovery after completing my Ph.D.

EDUCATION

University at Buffalo

Ph.D. in Computational Chemistry

Expected February 2027

Gettysburg College

B.S. in Biochemistry & Molecular Biology

May 2021

INDUSTRY EXPERIENCE

Enveda

Platform Chemistry Intern

May 2025 – August 2025

Boulder, CO

Project: *ML- and MD-Guided Natural Product Prioritization for Drug Discovery*

Challenge: The structural complexity and chemical diversity of natural products make it difficult to efficiently identify promising compounds from multi-million-member chemical libraries for experimental evaluation.

Action: Built and deployed computational pipelines integrating machine-learning models, molecular dynamics simulations, and virtual screening to predict compound bioactivity, prioritize natural products, and provide structural insights for medicinal chemistry teams.

Results:

- Prioritized compounds with predicted bioactivity for experimental screening, accelerating candidate selection from large chemical libraries.
- Delivered ML- and simulation-driven analyses that directly informed hit identification and early lead-optimization decisions.
- Collaborated with medicinal chemists and computational scientists to translate model predictions into actionable drug-discovery hypotheses.

ACADEMIC EXPERIENCE

Rockefeller University

Summer Research Fellow

May 2026 – Present

New York, NY

Project: *Bayesian Reweighting of RNA Structural Ensembles Using Cryo-EM Particle Images*

Challenge: RNA molecules are intrinsically dynamic, and a single static structure provides an incomplete representation of their conformational behavior. Although cryo-EM particle images contain information about structural heterogeneity and conformational populations, conventional reconstruction methods often average this information or divide it into a limited number of discrete classes.

Action: Developed an end-to-end computational workflow integrating molecular simulation, structural clustering, cryo-EM image simulation, deep-learning-based likelihood estimation, and Bayesian inference to determine RNA conformational populations directly from experimental particle images.

Results:

- Constructed representative RNA structural ensembles by clustering simulated conformations, extracting cluster-center structures, and quantifying conformational diversity through pairwise structural comparisons.
- Developed a GPU-accelerated training and inference pipeline for analyzing approximately 1.2 million experimental cryo-EM particle images, including image preprocessing, synthetic particle generation, classifier training, and fine-tuning.
- Integrated particle-level likelihood predictions with Bayesian population-calibration methods to estimate the relative populations of RNA conformational states.
- Implemented quantitative diagnostics for training convergence, class-specific accuracy, prediction entropy, and systematic class bias, identifying failure modes and guiding optimization of the structural ensemble and model-training strategy.

University at Buffalo

Computational Chemistry Doctoral Candidate

February 2021 – Present

Buffalo, NY

Project: *Multiscale Modeling of RNA and Membrane-Protein Structure–Property Relationships*

Challenge: Biomolecular function emerges from dynamic and heterogeneous structural ensembles that are difficult to characterize using static experimental structures alone. Understanding these systems requires physically accurate models, large-scale simulations, and quantitative methods for connecting molecular interactions to experimentally observable properties.

Action: Developed and validated atomistic and coarse-grained models of oligonucleotides and transmembrane proteins. Conducted GPU-accelerated molecular dynamics simulations on cloud and institutional high-performance computing platforms and built Python-based workflows for structural, thermodynamic, and dynamical analysis.

Results:

- Characterized how RNA sequence, temperature, and ionic conditions influence base stacking, chain compaction, conformational heterogeneity, and phase behavior.
- Quantified ion-mediated interactions, molecular organization, and chain dynamics within RNA condensates using custom structural and statistical analysis workflows.
- Extracted physicochemical descriptors from large molecular ensembles and applied dimensionality reduction and clustering to identify representative conformational states.
- Developed reproducible GPU-accelerated simulation and analysis pipelines deployed across Microsoft Azure, Google Cloud, and academic high-performance computing systems.
- Communicated computational findings in peer-reviewed publications, scientific presentations, and open-source research software.

University at Buffalo

Chemical Biology Research Assistant

September 2021 – January 2023

Buffalo, NY

Project: *Development of Bioorthogonal Probes for Protein and Live-Cell Labeling*

Challenge: Selective labeling of proteins in complex biological environments requires bioorthogonal probes with high reactivity, chemical stability, and minimal interference with native cellular processes.

Action: Designed and synthesized novel bioorthogonal probes and characterized their identity and purity using NMR and mass spectrometry. Evaluated reaction kinetics and stability using UV–visible spectroscopy and mass spectrometry, incorporated probe-conjugated unnatural amino acids into proteins, and maintained mammalian cell cultures for live-cell labeling experiments.

Results:

- Established a synthesis and analytical characterization workflow for evaluating the purity, stability, and reactivity of candidate bioorthogonal probes.
- Quantified the incorporation efficiency of probe-conjugated unnatural amino acids in proteins using mass spectrometry.

- Supported the application of newly developed probes from small-molecule characterization to protein-labeling and live-cell assay applications.

TECHNICAL SKILLS

Computational Chemistry & Biophysics: Atomistic and coarse-grained molecular dynamics • Force-field development • Free-energy perturbation • RNA and protein modeling • Structural ensemble analysis • Structure–property relationship analysis

Drug Discovery & Cheminformatics: Molecular docking • Virtual screening • Molecular descriptors • Molecular property prediction

Machine Learning & Data Analysis: XGBoost • Chemprop • PyTorch • Bayesian inference • Dimensionality reduction • Clustering • RDKit

Molecular Modeling & Structural Biology Software: Schrödinger Suite (Glide, FEP+) • OpenMM • GROMACS • AMBER • CHARMM • LAMMPS • AutoDock • AlphaFold • Boltz • CRY SOL

Programming & Scientific Computing: Python • Bash • C++ • SQL • Linux • HPC • GPU-accelerated computing • Docker • Git/GitHub • Jupyter • Globus

PUBLICATIONS

Research Articles

6. **Zhang, H.**; Baidya, L.; Nguyen, H. T. Temperature-dependent ion partitioning remodels RNA structure and internal organization in condensate. *Submitted*, 2026.
5. Mohanta, D.; **Zhang, H.**; Thirumalai, D.; Nguyen, H. T. Molecular Origins of Heterogeneous Aging and Spatial Organization in RNA Condensates. *Under Review*, 2026.
4. **Zhang, H.**; Maity, H.; Nguyen, H. T. Temperature-Dependent Ion Migration Underlies Sequence-Specific RNA Collapse. *Biophys. J.* **2026**, *125* (13), 3456–3470.
3. Baidya, L.; **Zhang, H.**; Nguyen, H. T. Poly(ADP-Ribose) (PAR) Exhibits Ion-Dependent Structural Properties Distinct from RNA. *Nucleic Acids Res.* **2026**, *54* (6), gkag265.
2. Maity, H.; **Zhang, H.**; Thirumalai, D.; Nguyen, H. T. RNA Structural Complexity Dictates Its Ion Atmosphere. *J. Phys. Chem. Lett.* **2025**, 8393–8402.
1. Fang, M.; Kumar, G. S.; Racioppi, S.; **Zhang, H.**; Rabb, J. D.; Zurek, E.; Lin, Q. Hydrazonyl Sultones as Stable Tautomers of Highly Reactive Nitrile Imines for Fast Bioorthogonal Ligation Reaction. *J. Am. Chem. Soc.* **2023**, *145* (18), 9959–9964.

Review Articles

3. **Zhang, H.**; Nguyen, H. T. Multiscale Models of Nucleic Acid–Driven Assembly and Biomolecular Condensation. *Manuscript in preparation*, 2026.
2. Abidakun, A. O.; **Zhang, H.**; Nguyen, H. T. Molecular Insights into the Regulation of RNA Conformational Transitions : Ensembles, Dynamics, and Cellular Heterogeneity. *Manuscript in preparation*, 2026.
1. **Zhang, H.**; Fang, M.; Lin, Q. Photo-Activatable Reagents for Bioorthogonal Ligation Reactions. *Top. Curr. Chem.* **2023**, *382* (1), 1.

PRESENTATIONS

10. **Zhang, H.**; Nguyen, H.T. RNA collapse and condensate formation tuned by temperature and counterions. *Poster Presentation. Biophysical Society Meeting, San Francisco, CA, Feb 2026.*

9. **Zhang, H.**; Nguyen, H.T. Coarse-grained Simulation of ssRNA. *Oral Presentation. MDAnalysis UGM, Tucson, AZ, Nov 2025.*
8. **Zhang, H.**; Nguyen, H.T. Temperature-Dependent Ion Migration Underlies Sequence-Specific RNA Collapse. *Poster Presentation. Rustbelt RNA Conference, Huron, Ohio, Oct 2025.*
7. **Zhang, H.**; Nguyen, H.T. Ion-dependent Structural Ensemble and Phase Separation Propensity of Single-stranded RNA. *Poster Presentation. Academic Excellence Celebration, Buffalo, NY, May 2025.*
6. **Zhang, H.**; Nguyen, H.T. Coarse-grained Simulation of RNA. *Invited Lecture. Molecular Dynamics Simulations by Prof. Viviana Monje, Buffalo, NY, April 2025.*
5. **Zhang, H.**; Nguyen, H.T. Ion-dependent Structural Ensembles and Phase Separation of Single-stranded RNA. *Oral Presentation. Buffalo RNA Group, Buffalo, NY, March 2025.*
4. **Zhang, H.**; Nguyen, H.T. Ion-dependent Structural Ensembles and Phase Separation of Single-stranded RNA. *Poster Presentation. Biophysical Society Meeting, Los Angeles, CA, Feb 2025.*
3. **Zhang, H.**; Nguyen, H.T. Ion-driven RNA Condensate Simulation. *Poster Presentation. Biophysical Society Meeting, Los Angeles, CA, Feb 2025.*
2. **Zhang, H.**; Nguyen, H.T. Ion-dependent Structural Ensemble of Single-stranded RNA. *Poster Presentation. New York State RNA Conference, Canandaigua, NY, Oct 2024. (Best Poster Award.)*
1. **Zhang, H.**; Nguyen, H.T. Parameterization of Coarse-grained ssRNA Force Field. *Poster Presentation. Buffalo Graduate Student Symposium, Buffalo, NY, May 2024.*

TEACHING

Computational Chemistry	<i>Spring 2025</i>
Physical Chemistry	<i>Spring 2022 and 2023, Fall 2024</i>
Honors Organic Chemistry	<i>Fall 2022</i>
General Chemistry	<i>Fall 2021 and 2025, Spring 2026</i>

SERVICES

Graduate Student Chemistry Club Treasurer	<i>Sep 2024 to Sep 2025</i>
Department Merchandise Designer and Purchaser	<i>March 2025 to Sep 2025</i>
Lab Safety Officer	<i>Jan 2022 to June 2023</i>
Lab Supplies Manager	<i>Jan 2022 to June 2023</i>
Diversity and Inclusion Committee	<i>Sep 2021 to Dec 2021</i>

AWARDS

Enveda Employee of the Month	<i>Aug 2025</i>
<i>Affiliation: Enveda</i>	<i>Boulder, CO</i>

- Description: Nominated by peers for the person most exemplifying Enveda's core values of curiosity, agency, journey, charity, and unity.

Academic Excellence Award	<i>May 2025</i>
<i>Affiliation: University at Buffalo</i>	<i>Buffalo, NY</i>

- Description: Recognize students for outstanding achievements throughout their SUNY experience in areas including academics, leadership, campus involvement, community service, and the arts.

New York RNA Symposium Certificate of Excellence

Oct 2024

Affiliation: The RNA Institute

Albany, NY

- Description: Peter Zhang has been awarded an RNA Society membership for their poster at the 2024 New York RNA Symposium, October 14-15, 2024. Organized by the Center of Excellence in RNA Research & Therapeutics. Honored by the University of Rochester and University at Albany.

John B. Zinn Chemistry Research Award

May 2021

Affiliation: Gettysburg College

Gettysburg, PA

- Description: Created by Frances and John Zinn in honor of John B. Zinn '09, who was professor of chemistry at the College from 1924-1959. Awarded to the senior making the greatest contributions in their own research in chemistry and to the research activities of the Department of Chemistry.